Week 6: The Confidence Machine

Probability & Bayes' Rule

FIT1061: Introduction to Artificial Intelligence

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The Story: Watson on *Jeopardy!*

On 14 February 2011, a computer built by IBM walked onto the set of an American television quiz show and, over three nights, beat the two greatest human champions the show had ever produced. The machine was called **Watson**. The champions were **Ken Jennings**, who once won 74 games in a row, and **Brad Rutter**, who had never lost a match to a human. Watson beat them both. To understand why that mattered (and why it is the perfect way into this week) you first have to understand the game.

Why the game is hard for a machine

Jeopardy! runs backwards. Instead of asking a question and waiting for an answer, the show gives you the **answer** and asks you to supply the **question**. A clue might read, *"This 'father of our country' didn't really chop down a cherry tree,"* and the correct response is *"Who is George Washington?"* That inversion is the least of the difficulties. The clues are dense with **puns, wordplay, and deliberate ambiguity**. They range across **any topic at all** (history, science, pop music, cooking, sport) so there is no small, tidy body of knowledge to master. And the game is played at **buzzer speed**: three contestants race to press a button, and only the fastest correct responder scores. A player must not only *know* the answer but be confident enough to commit to it in a fraction of a second, because a wrong buzz loses points.

For a machine, every one of those features is a wall. Read a punning clue and you must work out what **kind** of thing is even being asked for: a person? a city? a year? Then you must search an enormous store of documents for candidate answers, weigh them, and decide whether you are sure enough to buzz, all faster than a human champion who has spent a lifetime training on exactly this. This is open-domain language, under time pressure: one of the hardest problems in the field.

How Watson actually worked

Here is the crucial detail, and it is the seed of everything this week. Watson **did not know things** the way a person does. For each clue, it ran a pipeline that looked roughly like this:

- It received the clue as **text**: typed in electronically, at the same moment the humans saw it. Watson had **no speech recognition**; the friendly synthesised voice and the glowing on-screen avatar were stagecraft, not the actual system listening.
- It parsed the clue, guessed the expected **answer type**, and searched millions of ingested documents in parallel.
- Out of that search it generated **hundreds of candidate answers**: not one guess, but a whole ranked field of possibilities.
- It assigned each candidate a **confidence score**: a number capturing how strongly the evidence pointed that way.
- It **buzzed in only when the top candidate's confidence rose above a threshold.** Below the line, it stayed silent rather than risk a costly wrong answer.

The machine had been trained on a goldmine: fans of the show had archived roughly **100,000 past clues and their answers**, giving IBM a vast set of labelled examples to tune the system on.

Notice what this means. Watson was **never certain.** It was, at best, **probably right**, and it had a principled way of deciding when "probably" was good enough to act on. The whole apparatus was a machine for turning messy evidence into a number between 0 and 1, and then acting on that number.

That it was pattern-matching rather than understanding showed in its mistakes. In a category called *U.S. Cities*, Watson once answered **"Toronto"**: a Canadian city. On a clue about an anatomical oddity, it replied **"leg"** rather than **"a missing leg."** The system had no common-sense grip that a missing leg is stranger than a leg; it was matching text, scoring candidates, and sometimes the score pointed the wrong way. Impressive and genuinely useful, but confident is not the same as correct.

That gap (between a high confidence score and the truth) is exactly what this week gives you the tools to reason about. **How does a machine act on beliefs it is not certain about?** The answer is **probability**, and its single most important tool: **Bayes' rule.**

Probability by Counting

Before any formulas, the reassuring news: **probability begins as counting.** If you can build a table of counts and divide, you can compute every probability we need this week.

A small, concrete world

Imagine an inbox of just **10 emails**. Some are **spam**; the rest are ordinary mail, which we will call **ham**. Independently, some emails contain the word **"FREE"** and some do not. Suppose we go through all ten by hand and tally them into a table:

	FREE	no FREE	**total**
spam	3	1	**4**
ham	2	4	**6**
total	**5**	**5**	**10**

Every number we could want about this inbox is already sitting in this table. Reading it off is a matter of picking the right cell, row, or column and dividing by the total.

Turning counts into probabilities

A **probability** is just a count divided by the total number of possibilities. Divide each count in the table by 10:

What we ask	Count	Probability
It is spam	4 / 10	$P(\text{spam}) = 0.40$
It contains FREE	5 / 10	$P(\text{FREE}) = 0.50$
It is spam **and** contains FREE	3 / 10	$P(\text{spam} \cap \text{FREE}) = 0.30$

Two names are worth learning here. A **marginal** probability comes from a whole row or column total: $P(\text{spam}) = 0.40$ reads off the "spam" row total of 4. A **joint** probability comes from a single inner cell: $P(\text{spam} \cap \text{FREE}) = 0.30$ reads off the one cell where both conditions hold. The symbol $\cap$ simply means **"and"**: both things true at the same time.

> **!** Key idea: every probability is a piece of one table
>
> A single count, divided by the total, is a probability. Marginals come from the row and column totals; joints come from the inner cells. There is no new machinery: just one count table, read in different ways.

Conditional Probability

Now the question that actually matters for spam filtering. A **new** email arrives, and it contains the word **"FREE"**. Given only that fact, how likely is it to be spam? We are no longer asking about all ten emails; we are asking about the *subset* that contains FREE.

Narrowing the table

The intuition is to **narrow the world down** to the evidence we have, then look at the fraction we care about inside it. We saw FREE, so ignore the "no FREE" column entirely and keep only the FREE column:

	FREE
spam	3
ham	2
total	**5**

Inside this narrowed world there are **5** emails. Of those, **3** are spam. So the chance our new email is spam is:

$$P(\text{spam} \mid \text{FREE}) = \frac{3}{5} = 0.60.$$

The vertical bar | is read **"given"**: $P(\text{spam} \mid \text{FREE})$ is the probability of spam *given that* we have seen FREE. Notice that seeing FREE **shifted our belief**: before we knew anything, an email was spam with probability 0.40; once we learned it contained FREE, that rose to 0.60. **Evidence updates belief.**

The formula, and why it matches

"Narrowing the table" has a formal name and a formula:

> **Definition: conditional probability**
>
> For any two events A and B (with $P(B) > 0$),
>
> $$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$
>
> Narrow the world down to B, then take the fraction that is *also* in A.

Let us check it gives the same answer. We already read off $P(\text{spam} \cap \text{FREE}) = 0.30$ and $P(\text{FREE}) = 0.50$ from the full table. So

$$P(\text{spam} \mid \text{FREE}) = \frac{P(\text{spam} \cap \text{FREE})}{P(\text{FREE})} = \frac{0.30}{0.50} = 0.60.$$

The two routes **agree**, and it is worth seeing *why* they must. Dividing the joint by the marginal, $\frac{0.30}{0.50}$, is the same arithmetic as dividing the joint **count** by the marginal **count**, $\frac{3}{5}$: the "divide by 10" in the numerator and denominator simply cancels. "Narrow the FREE column" and "apply the formula" are two descriptions of one calculation.

Bayes' Rule

We can now flip a conditional, and flipping is the key move in all of probabilistic AI.

Two different questions

These two questions look similar but are **not** the same:

- $P(\text{spam} \mid \text{FREE})$: *given* that an email contains FREE, is **this** email spam? This is what we actually **want**: we are handed a new email, we see the word FREE, and we must classify it. Reasoning from evidence back to a cause is the **diagnostic** direction.
- $P(\text{FREE} \mid \text{spam})$: *given* that an email is spam, how often does it contain FREE? This is what we can most easily **count** from labelled training data: take all the known spam, and see what fraction says FREE. Reasoning from a cause forward to its evidence is the **causal** direction.

In general these two are different numbers. From our table, $P(\text{spam} \mid \text{FREE}) = \frac{3}{5} = 0.60$, while $P(\text{FREE} \mid \text{spam}) = \frac{3}{4} = 0.75$. **Bayes' rule** is the bridge that lets us compute the one we want from the one we can count.

> **Definition: Bayes' rule**
>
> $$P(H \mid D) = \frac{P(D \mid H)\, P(H)}{P(D)}.$$
>
> Here H is the **hypothesis** (the email is spam) and D is the **data** we observed (it contains FREE). This is the same conditional-probability formula, rearranged, with $A = H$ and $B = D$.

Each piece has a name worth memorising:

Part	Symbol	Meaning	Our value
prior	$P(H)$	belief *before* seeing evidence	0.40
likelihood	$P(D \mid H)$	how often the hypothesis produces the data	0.75
evidence	$P(D)$	how common the data is overall	0.50
posterior	$P(H \mid D)$	belief *after* seeing evidence	?

Working it through

Take each part straight from the count table:

- **likelihood** $P(\text{FREE} \mid \text{spam}) = \frac{3}{4} = 0.75$: of the 4 spam emails, 3 say FREE.
- **prior** $P(\text{spam}) = \frac{4}{10} = 0.40$: 4 of the 10 emails are spam.
- **evidence** $P(\text{FREE}) = \frac{5}{10} = 0.50$: 5 of the 10 emails say FREE.

Substitute into Bayes' rule:

$$P(\text{spam} \mid \text{FREE}) = \frac{P(\text{FREE} \mid \text{spam})\,P(\text{spam})}{P(\text{FREE})} = \frac{(0.75)(0.40)}{0.50} = \frac{0.30}{0.50} = 0.60.$$

The same 0.60 we got by narrowing the column. That is not a coincidence: the numerator $P(\text{FREE} \mid \text{spam})\,P(\text{spam}) = (0.75)(0.40) = 0.30$ is exactly the joint probability $P(\text{spam} \cap \text{FREE})$, rebuilt from the causal direction. Bayes' rule is the joint-count table, rearranged.

Where the evidence comes from

Often we do not know $P(D)$ directly, but we can build it. The **law of total probability** says the data can arrive through the hypothesis being true *or* being false, so add up both routes:

$$P(D) = P(D \mid H)\,P(H) + P(D \mid \neg H)\,P(\neg H),$$

where $\neg H$ means "not H": the ham case here. For our inbox, $P(\text{FREE} \mid \text{ham}) = \frac{2}{6} = \frac{1}{3}$ (of the 6 ham emails, 2 say FREE), so

$$P(\text{FREE}) = (0.75)(0.40) + \left(\tfrac{1}{3}\right)(0.60) = 0.30 + 0.20 = 0.50.$$

Exactly the 0.50 we read off the table. The law of total probability is just a careful way of counting *every* path to the data, and it is the piece we will lean on hardest in the puzzle below, where the evidence cannot be read off directly.

> **!** Key idea: Bayes is rearranged counting
>
> Prior, likelihood, and evidence all come from the one count table. Bayes' rule is not a new fact about the world: it is bookkeeping that lets us **flip** a conditional we can count into the conditional we actually want. **Count. Condition. Flip.**

Worked Example: The Base-Rate Trap

Here is a puzzle that fools most people, and famously most doctors.

> A disease affects **1%** of people. A test for it is **99% accurate**. You take the test and it comes back **positive**. What is the chance you actually have the disease?

"99% accurate" means the test is right 99% of the time on sick and healthy people alike: it correctly flags 99% of sick people, and correctly clears 99% of healthy people. Most people, hearing a 99% test come back positive, answer **about 99%**. The real answer is **about 50%**. To see why, we count.

Imagine 10,000 people

Pick a concrete population (10,000 people) and follow the two groups through the test:

Group	How many	Test **positive**	Test **negative**
Sick (1% of 10,000)	100	99	1
Healthy (99% of 10,000)	9,900	99	9,801
Total	10,000	**198**	9,802

Read the two "positive" figures carefully. Among the **100 sick** people, the 99%-accurate test correctly flags **99** of them. But among the **9,900 healthy** people, the test is wrong 1% of the time, and 1% of 9,900 is **99** healthy people who test positive by mistake (these are **false positives**). So **198** people test positive in total, and only **99** of them are actually sick.

The reveal

The chance you are sick, given a positive test, is therefore

$$P(\text{sick} \mid \text{positive}) = \frac{99}{99 + 99} = \frac{99}{198} = 0.50.$$

The very same answer falls out of Bayes' rule directly. First the evidence, by the law of total probability:

$$P(\text{positive}) = \underbrace{(0.99)(0.01)}_{\text{sick}\to+} + \underbrace{(0.01)(0.99)}_{\text{healthy}\to+} = 0.0099 + 0.0099 = 0.0198.$$

Then flip:

$$P(\text{sick} \mid \text{positive}) = \frac{P(\text{positive} \mid \text{sick})\, P(\text{sick})}{P(\text{positive})} = \frac{(0.99)(0.01)}{0.0198} = 0.50.$$

Why the base rate wins

The answer feels wrong until you see what is fighting what. The disease is **rare**: its **base rate**, how common it is before any test, is just 1%. That tiny sick pool produces only 99 true positives. The healthy pool is **99 times larger**, so even a small 1% error rate turns it into an equally large flood of 99 false positives. The two piles are the same height, and a positive result is a coin toss between them. A 99% likelihood collapsed into a 50% posterior, and if the disease were rarer still, a positive result would more often be *wrong* than right.

> **!** Key idea: rare conditions make a positive result hard to trust
>
> When the base rate is tiny, even a highly accurate test produces about as many false positives as true ones, because it is applied to a vast pool of people who do not have the condition. The prevalence of the condition matters as much as the accuracy of the test. Ignore the base rate and you will badly overestimate what a positive result means.

This is the deeper lesson behind Watson's confidence score. A number close to certainty can still be misleading if the thing you are looking for is rare. Confidence is not correctness.

Summary

Key takeaways:

- **Probability is counting.** Build a count table, divide by the total, and you have every marginal, joint, and conditional probability you need.
- **Conditional probability narrows the world.** $P(A \mid B) = P(A \cap B)/P(B)$: restrict to the evidence B, then take the fraction also in A.
- **Bayes' rule flips the conditional.** We usually *want* $P(H \mid D)$ (diagnostic) but can only *count* $P(D \mid H)$ (causal). Bayes bridges them: $P(H \mid D) = P(D \mid H)\,P(H)/P(D)$.
- **The evidence** $P(D)$ is built with the law of total probability: add up every path to the data.
- **The base rate dominates.** A 99%-accurate test on a 1%-rare disease is only 50% reliable when it fires. Confidence is not the same as correctness.

Key vocabulary:

Term	Meaning
Conditional probability	The probability of one event *given* that another has occurred; found by narrowing to the evidence
Prior $P(H)$	Belief in the hypothesis *before* seeing any evidence
Likelihood $P(D \mid H)$	How often the hypothesis produces the observed data: the direction we can count
Evidence $P(D)$	How common the observed data is overall, across all hypotheses
Posterior $P(H \mid D)$	Belief in the hypothesis *after* updating on the evidence: the direction we want
Base rate	How common a condition is before any test or evidence is considered
False positive	A test result that says "yes" when the truth is "no"

What's next: we can flip the conditional for **one** word of evidence, like FREE. But real messages contain **many** words at once. Next we meet the shortcut that makes reasoning over many pieces of evidence at once manageable.

In the Lab

This week: counting your way from raw emails to a calibrated belief.

You will put every idea above into working code, starting from a small set of labelled emails.

1. **Build the count table.** Take a collection of emails, each labelled spam or ham and each marked for whether it contains a chosen word, and tally them into the 2×2 table of counts.
2. **Compute a conditional probability** by narrowing the table: restrict to the emails containing the word, and read off the fraction that are spam.
3. **Apply Bayes' rule** to flip the conditional: turn the likelihood you can count, $P(\text{word} \mid \text{spam})$, together with the prior and the evidence, into the posterior $P(\text{spam} \mid \text{word})$ you actually want. Confirm it matches the narrowed-table answer.
4. **Hand-compute the base-rate posterior.** Given your own choice of prevalence and test accuracy, work out the chance of really being sick after a positive result, and see for yourself how the base rate reshapes the answer.

Everything you need is the count table and division. There is no new mathematics beyond what is in these lecture notes; only careful counting.